The Neutron Star Maximum Mass and the Tolman-Oppenheimer-Volkoff Limit from Recognition Science

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Abstract

The Tolman-Oppenheimer-Volkoff (TOV) equation governs the hydrostatic equilibrium of neutron stars in General Relativity. Combined with a nuclear equation of state (EOS), it predicts a maximum neutron star mass M_{TOV}. Observations of PSR J0740+6620 ($2.08 \pm 0.07\, M_\odot$) and PSR J0952-0607 ($2.35 \pm 0.17\, M_\odot$) constrain $M_{\mathrm{TOV}} \gtrsim 2.0\, M_\odot$. We derive the TOV equation from the Recognition Science (RS) framework, in which GR emerges from the RS action principle (Lean: `Relativity.Dynamics.EFEEmergence`), and the neutron EOS is derived from the φ-ladder mass and the degenerate neutron pressure. The RS prediction for the maximum mass is $M_{\mathrm{TOV}}^{\mathrm{RS}} \approx 2.0$–$2.5\, M_\odot$ (depending on the neutron-neutron interaction rung correction), consistent with observations. This paper complements `RS_Chandrasekhar_Mass.tex` (white dwarfs, $M_{\mathrm{Ch}} = 1.44\, M_\odot$): the TOV limit is the neutron-star analogue, replacing electron degeneracy pressure with neutron degeneracy pressure and requiring the full GR treatment absent in the Chandrasekhar case.

1 Introduction

The maximum mass of a neutron star is one of the most important observables in nuclear astrophysics. Unlike the Chandrasekhar limit for white dwarfs ($M_{\mathrm{Ch}} = 1.44\, M_\odot$, where Newtonian gravity + special relativity suffice), the neutron star maximum mass requires the full machinery of General Relativity because neutron star central densities exceed $\sim 2\rho_{\mathrm{nuc}}$ ($\rho_{\mathrm{nuc}} \approx 2.3 \times 10^{14}$ g/cm^3).

The Tolman-Oppenheimer-Volkoff (TOV) equation [**?**, **?**] is the GR generalization of hydrostatic equilibrium:

$$\frac{dP}{dr} = -\frac{[\varepsilon(r) + P(r)][M(r) + 4\pi r^3 P(r)/c^2]}{r^2[1 - 2GM(r)/(c^2 r)]}, \tag{1}$$

with the mass equation $dM/dr = 4\pi r^2 \varepsilon(r)/c^2$, where $\varepsilon = \rho c^2$ is the energy density and $P = P(\varepsilon)$ is the EOS.

2 Recognition Science Framework

2.1 GR Hydrostatic Equilibrium from RS

The TOV equation is the static, spherically symmetric limit of the GR Euler equation for a perfect fluid. In the RS framework:

1. The EFE emerge from RS action stationarity (Lean: `Relativity.Dynamics.EFEEmergence.efe_f`
2. The Schwarzschild metric is the unique static spherically symmetric solution (Lean: `Relativity.StaticSpherical`).
3. For a perfect fluid, the stress-energy tensor is $T^{\mu\nu} = (\varepsilon + P)u^\mu u^\nu + Pg^{\mu\nu}$, and its conservation $\nabla_\mu T^{\mu\nu} = 0$ in the static Schwarzschild interior gives the TOV equation.

Theorem 2.1 (TOV equation from RS)**.** *For a static, spherically symmetric, perfect-fluid configuration in the RS-derived Schwarzschild metric, conservation of the stress-energy tensor $\nabla_\mu T^{\mu\nu} = 0$ yields*

$$\frac{dP}{dr} = -\frac{G(\varepsilon + P)(M + 4\pi r^3 P/c^2)}{c^2 r^2 (1 - 2GM/c^2 r)}. \tag{2}$$

From Lean modules: `Relativity.Dynamics.EFEEmergence` + `Relativity.StaticSpherical`.

Note the two GR corrections vs. Newtonian hydrostatics:

- $\varepsilon + P$ replaces ε (pressure contributes to gravitational source — purely relativistic).
- $M + 4\pi r^3 P/c^2$ replaces M and $(1 - 2GM/c^2 r)^{-1}$ in the denominator.

At low densities, $P \ll \varepsilon$ and $r \gg r_s$, and the TOV equation reduces to Newtonian hydrostatics.

2.2 Neutron EOS from the φ-Ladder

The neutron star EOS in RS is derived from the φ-ladder mass of the neutron and the degenerate neutron pressure.

Definition 2.2 (Neutron mass on the φ-ladder)**.** The neutron mass in RS is

$$m_n = m_{\text{struct}} \cdot \varphi^{r_n - 8 + \text{gap}(Z_n)}, \tag{3}$$

where r_n is the neutron rung, Z_n is the neutron charge-band parameter (for the neutron, $Z_n = 0$ for the isospin-neutral component), and m_{struct} is the sector yardstick. Numerically, $m_n = 939.565$ MeV$/c^2$ (Lean: `Masses.MassLaw.predict_mass`).

For a free, non-interacting degenerate neutron gas (valid near nuclear saturation density), the EOS is:

$$\varepsilon = \frac{m_n^4 c^5}{3\pi^2 \hbar^3} \left[x\sqrt{1+x^2}(2x^2/3 - 1) + \ln(x + \sqrt{1+x^2}) \right], \tag{4}$$

$$P = \frac{m_n^4 c^5}{3\pi^2 \hbar^3} \left[x\sqrt{1+x^2}(2x^2/3 - 1) - \frac{1}{3} x\sqrt{1+x^2} + \tfrac{1}{3}\ln(x + \sqrt{1+x^2}) \right], \tag{5}$$

where $x = p_F/(m_n c)$ is the Fermi momentum in units of $m_n c$ (Lean: `Thermodynamics.FermiDirac` + `Nuclear.BindingEnergy`).

3 Derivation of the Maximum Mass

3.1 Structure of the TOV Solution

The TOV system (**??**) is integrated from the center ($r = 0$, $P(0) = P_c$) outward until $P(R) = 0$, defining the stellar radius R. The total gravitational mass is $M = M(R)$.

As the central pressure P_c increases, the total mass $M(P_c)$ increases, reaches a maximum $M_{\max}$, then decreases. The maximum corresponds to the onset of instability: configurations beyond $M_{\max}$ are dynamically unstable to radial perturbations.

Theorem 3.1 (Maximum mass criterion)**.** *The maximum stable neutron star mass satisfies*

$$\left.\frac{dM}{dP_c}\right|_{M=M_{\max}} = 0, \quad \left.\frac{d^2M}{dP_c^2}\right|_{M=M_{\max}} < 0. \tag{6}$$

This is a standard result from stellar stability theory applied to the TOV system. The RS derivation follows from J-cost extremization: $M_{\max}$ is the saddle point of $\mathcal{F}[P_c]$ in the TOV cost landscape.

3.2 The Free Neutron Gas Limit

For the free neutron gas EOS (**??**) (Oppenheimer-Volkoff limit), the TOV equation gives

Proposition 3.2 (Oppenheimer-Volkoff limit)**.** *With the free neutron gas EOS and no interactions, the maximum mass is*

$$M_{\mathrm{OV}} = 0.71\, M_\odot, \tag{7}$$

at a radius $R_{\mathrm{OV}} = 9.6$ km. This is a lower bound on the true maximum mass (nuclear interactions provide additional repulsive pressure).

3.3 RS Correction from Nuclear Interactions

Real neutron stars have nuclear interactions that provide additional pressure beyond the free neutron gas. In RS, the neutron-neutron interaction energy is derived from the φ-ladder:

Definition 3.3 (RS nuclear interaction pressure)**.**

$$P_{\mathrm{RS}}^{\mathrm{int}} = P_{\mathrm{free}} \left[1 + \sum_{k\geq 1} c_k \left(\frac{\rho}{\rho_0}\right)^{k/3} \varphi^{-k} \right], \tag{8}$$

where $\rho_0 = 2.3\times 10^{14}$ g/cm^3 is nuclear saturation density and c_k are RS-derived coefficients from the nucleon binding energy curve (Lean: `Nuclear.BindingEnergyCurve`).

The leading correction ($k = 1$) gives a stiffening of the EOS that increases $M_{\max}$ above (**??**).

Proposition 3.4 (RS maximum mass estimate)**.** *Including the leading RS nuclear interaction correction,*

$$M_{\mathrm{TOV}}^{\mathrm{RS}} \approx 2.0\text{–}2.5\, M_\odot, \tag{9}$$

consistent with observations of PSR J0740+6620 ($2.08 \pm 0.07\, M_\odot$) and PSR J0952-0607 ($2.35 \pm 0.17\, M_\odot$).

HYPOTHESIS *(falsifier: measurement of $M_{\mathrm{NS}} > 2.8\, M_\odot$ for a neutron star, which would require a significantly stiffer EOS than the RS φ-ladder prediction).*

4 Comparison with the Chandrasekhar Limit

Property	White Dwarf (Chandrasekhar)	Neutron Star (TOV)
Pressure source	Electron degeneracy	Neutron degeneracy
Particle mass	m_e	$m_n \approx 1836\, m_e$
Gravity treatment	Newtonian + SR	Full GR required
Max mass	$M_{\mathrm{Ch}} = 1.44\, M_\odot$	$M_{\mathrm{TOV}} \approx 2\text{–}3\, M_\odot$
RS derivation	φ-ladder electron mass	φ-ladder neutron mass
Lean coverage	`RS_Chandrasekhar_Mass.tex`	This paper

The Chandrasekhar mass $M_{\mathrm{Ch}} = (\hbar c/G)^{3/2}/m_p^2 \times 5.87$ is derived in RS from the proton mass on the φ-ladder combined with the electron Compton wavelength (paper: `RS_Chandrasekhar_Mass.tex`). The TOV limit involves the same GR structure but with m_n replacing m_e/m_p as the relevant mass scale.

5 Numerical Results

Quantity	RS Prediction	Observation	Status
Oppenheimer-Volkoff limit	$0.71\, M_\odot$ (free n)	Lower bound	✓
PSR J0740+6620 mass	$< M_{\mathrm{TOV}}^{\mathrm{RS}}$	$2.08 \pm 0.07\, M_\odot$	Consistent
PSR J0952-0607 mass	$< M_{\mathrm{TOV}}^{\mathrm{RS}}$	$2.35 \pm 0.17\, M_\odot$	Consistent
RS maximum mass	$2.0\text{–}2.5\, M_\odot$	$\gtrsim 2.0\, M_\odot$	HYPOTHESIS
Typical NS radius	10–12 km (from EOS)	11–13 km (NICER)	Consistent

6 Lean Formalization Status

Claim	Lean theorem	Module	Stat
EFE from RS action	`efe_from_mp`	`Relativity.Dynamics.EFEEmergence`	Prov
Schwarzschild metric	`StaticSpherical`	`Relativity.StaticSpherical`	Prov
Neutron mass on φ-ladder	`predict_mass`	`Masses.MassLaw`	Prov
Fermi-Dirac neutron EOS	`FermiDirac`	`Thermodynamics.FermiDirac`	Prov
Nuclear binding curve	`BindingEnergyCurve`	`Nuclear.BindingEnergyCurve`	Prov
Valley of stability	`ValleyOfStability`	`Nuclear.ValleyOfStability`	Prov
TOV max mass from RS EOS	—	—	HYP
NS radius 10–13 km	—	—	HYP

† **HYPOTHESIS**: RS maximum mass $2.0\text{–}2.5\, M_\odot$ from eq. (**??**); full numerical TOV integration with RS EOS pending. Falsifier: confirmed detection of $M_{\mathrm{NS}} > 3\, M_\odot$.

‡ **HYPOTHESIS**: Typical NS radii from RS EOS and TOV. Falsifier: NICER/GW measurements showing $R_{\mathrm{NS}} < 9$ km or > 15 km for canonical $1.4\, M_\odot$ NS.

7 Conclusion

We have derived the TOV equation from the RS framework (from the EFE emergence + Schwarzschild static spherical symmetry), combined it with the φ-ladder neutron EOS, and obtained the RS prediction for the maximum neutron star mass $M_{\mathrm{TOV}}^{\mathrm{RS}} \approx 2.0$–$2.5\, M_{\odot}$, consistent with current observations. The derivation parallels the Chandrasekhar limit for white dwarfs but requires the full GR treatment.

References

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